

Racial Disparities in Maternal Mortality and Hospital Care Access in the United States (2021–2023)

SIADS 593: Milestone I Project

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Abstract

Maternal mortality rates in the United States exhibit persistent racial disparities, with Black women experiencing substantially higher mortality rates than White women (Hoyert, 2021; Fleszar et al., 2023). This project investigates whether state-level hospital characteristics and maternal care access are associated with the magnitude of these disparities across U.S. states between 2021 and 2023. Using national mortality data from CDC WONDER and hospital infrastructure data from the CMS Hospital General Information files, we construct state-level disparity metrics and examine their relationship to hospital availability, emergency service coverage, and electronic health record (EHR) interoperability. Our findings highlight meaningful geographic variation in racial disparities and suggest that structural differences in hospital systems may be associated with observed inequities.

Keywords: maternal mortality, hospital care access, emergency services, racial disparities, healthcare, population health

1 Introduction

Maternal mortality is widely recognized as a critical indicator of population health and health system performance (WHO, 2019; MacDorman et al., 2016). Despite advances in medical technology and prenatal care, the United States continues to report higher maternal mortality rates than other high-income countries, with persistent and substantial racial disparities (Tikkanen et al., 2020; Hoyert, 2023). Black women in the United States experience maternal mortality rates several times higher than those of White women, a disparity that has remained consistent across decades (Hoyert, 2023; Fleszar et al., 2023).

Although national-level disparities are well documented, less is known about how these disparities vary across states and whether differences in hospital characteristics and maternal care infrastructure are associated with their magnitude. Existing research highlights the roles of structural racism, healthcare system bias, and differential quality of care (Howell, 2018; Petersen et al., 2019), yet comparatively limited work has examined how state-level hospital systems and institutional characteristics may shape these inequities. Because maternal care is delivered within localized hospital systems, variation in hospital access, emergency capacity, and quality reporting may contribute meaningfully to geographic differences in maternal outcomes.

1.1 Research Questions

The goal of this project is not to look at singular events where mothers died to find whether bias took place. Rather, it is to examine whether racial disparities in maternal mortality between Black and White women are associated with differences in hospital characteristics and maternal care access across U.S. states between 2021 and 2023. By integrating maternal mortality data with hospital and healthcare access data, we aim to identify state-level patterns, contextual differences, and outliers that cannot be observed using either dataset alone. This project will use descriptive and exploratory analyses to characterize structural healthcare conditions associated with maternal mortality disparities, without making causal claims.

We developed (3) research questions to help guide our analysis:

1. How do racial disparities in maternal mortality between Black and White women vary across U.S. states from 2021–2023?
2. Are state-level hospital characteristics (e.g., emergency services availability, EHR interoperability, hospital density) associated with the size of these disparities?
3. Do states with greater hospital access and infrastructure exhibit smaller racial gaps in maternal mortality?

2 Data

2.1 Maternal Mortality Data (Primary Dataset)

Maternal mortality data were obtained from the CDC WONDER “Final Multiple Cause of Death” database. Records were extracted at the state–year–race level using ICD-10 codes 000–099 (pregnancy, childbirth, and puerperium).

- Years: 2021–2023
- Geography: U.S. states
- Race categories: Black or African American, White (Single Race 6 classification)
- Measures: Number of deaths, population counts

Deaths and population values were aggregated across years to mitigate suppression effects common in rare-event data. Crude mortality rates per 100,000 were computed from aggregated counts.

Table 1: Data Extraction and Processing Workflow

Step	Dataset	Processing Action	Output
1	CDC WONDER	Restricted to 2021–2023; selected ICD-10 codes O00–O99	Filtered mortality data
2	CDC WONDER	Converted suppressed counts to NA; preserved zeros	Cleaned death counts
3	CDC WONDER	Aggregated deaths and population by state, race, and year	State-year totals
4	CDC WONDER	Calculated mortality rates per 100,000 population	Race-specific mortality rates
5	CDC WONDER	Computed disparity metrics (gap and ratio)	Disparity dataset
6	CMS Hospital Data	Selected October 2021–2023 snapshots	Harmonized annual hospital data
7	CMS Hospital Data	Normalized binary variables (Yes/No → 1/0)	Standardized indicators
8	CMS Hospital Data	Aggregated to state-year level (proportions and averages)	State-level hospital metrics
9	Combined	Merged mortality and hospital datasets by state and year	Final analysis dataset

Table 2: Key Derived Variables Used in Analysis

Variable	Formula / Definition	Interpretation
$rate_{Black}$	$(Deaths_{Black} / Population_{Black}) \times 100,000$	Black maternal mortality rate
$rate_{White}$	$(Deaths_{White} / Population_{White}) \times 100,000$	White maternal mortality rate
Gap (Black–White)	$rate_{Black} - rate_{White}$	Absolute disparity in mortality rates
Ratio (Black/White)	$rate_{Black} / rate_{White}$	Relative disparity (risk ratio)
% Emergency Hospitals	$\#Hospitals \text{ with Emergency Services} / \#Total \text{ Hospitals}$	Structural access proxy
% EHR Interoperable	$\#Hospitals \text{ with EHR Interoperability} / \#Total \text{ Hospitals}$	Digital infrastructure proxy
Average Hospital Rating	Mean CMS overall hospital rating by state	General hospital quality indicator

2.2 Hospital Characteristics Data (Secondary Dataset)

Hospital data were drawn from the CMS Hospital General Information files (October releases) for 2021, 2022, and 2023.

- Unit of analysis: Individual hospitals
- Key variables:
 - Hospital ownership
 - Hospital type
 - Emergency services availability
 - EHR interoperability
- Temporal approach: October snapshots were selected as a consistent annual reference point to avoid seasonal duplication and reporting inconsistencies.

Hospital records were aggregated to the state level to derive proportions and counts (e.g., proportion of hospitals offering emergency services).

3 Methods

3.1 Data Cleaning and Preparation

Maternal Mortality Data

- Suppressed death counts were retained as missing values rather than zeros.
- Deaths and populations were aggregated by state and race across years (2021–2023).
- Crude mortality rates were calculated post-aggregation.
- States lacking complete data for both Black and White populations were excluded from disparity calculations.

Hospital Data

- Hospital datasets from 2021–2023 were cleaned and harmonized.
- NaN values in EHR interoperability were converted to 0s as the data dictionary suggested missing values were “no”s
- Binary indicators (e.g., emergency services, EHR interoperability) were normalized to consistent encodings (0: No, 1: Yes).
- Records were aggregated to the state level, producing:
 - Total hospital counts
 - Proportion of hospitals with emergency services
 - Proportion with EHR interoperability
- Only October snapshots were used to ensure consistency across years.

3.2 Disparity Metrics

Two complementary disparity metrics were computed:

1. Absolute gap:

$$Gap = Rate_{Black} - Rate_{White} \quad (1)$$

This metric represents the absolute excess maternal deaths per 100,000 population among Black women relative to White women.

2. Relative ratio:

$$Ratio = \frac{Rate_{Black}}{Rate_{White}} \quad (2)$$

This metric captures proportional inequity in maternal mortality rates.

Together, these metrics capture both magnitude and proportional inequity.

3.3 Hospital Metrics

Two hospital metrics were computed:

1. Emergency services ratio:

$$Ratio = \frac{Sum_{EmergencyServices}}{HospitalCount} \quad (3)$$

This metric captures the proportion of emergency services per hospital in a state by finding the sum of all hospitals with emergency services in proportion to all hospitals in any specific state.

2. EHR Interoperability:

$$Ratio = \frac{Sum_{EHRInteroperability}}{HospitalCount} \quad (4)$$

This metric captures the proportion of EHR Interoperability per hospital in a state by finding the sum of all hospitals with EHR Interoperability in proportion to all hospitals in any specific state.

Together these statistics are used to approximate availability, and functionality of hospitals in a given state.

3.4 Analytical Approach

- Descriptive statistics summarized national and state-level distributions.
- Choropleth maps visualized geographic variation.
- Correlation analyses explored associations between hospital characteristics and disparity metrics.

4 Results

4.1 National Overview of Racial Disparities

To better understand the geographic distribution and magnitude of racial disparities in maternal mortality, we ranked states annually based on the absolute Black–White maternal mortality gap (per 100,000 live births). The ranked bar charts highlight states with the lowest and highest disparities for 2021, 2022, and 2023.

4.1.1 2021 Results

In 2021, substantial variation was observed across states. The largest disparities were concentrated in Southern and Midwestern states. Florida (1.3), Ohio (1.2), Mississippi (1.2), Tennessee (1.1), and Louisiana (1.1) exhibited the highest gaps.

Conversely, New Mexico (-0.8) and Minnesota (-0.3) showed negative disparities, indicating either parity or slightly higher mortality rates among White mothers relative to Black mothers. Vermont and Hawaii reported gaps of approximately zero.

Key observations for 2021:

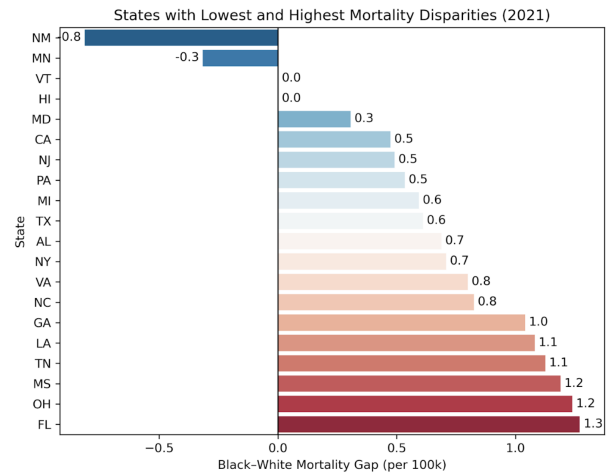


Figure 1: Ranked state-level Black–White maternal mortality gaps (2021), highlighting states with lowest and highest disparities.

- Clear regional clustering in the Southeast.
- Several states exhibited gaps greater than 1.0 per 100,000.
- The presence of negative or zero values suggests variability in population size, reporting practices, or demographic structure.

The 2021 rankings indicate that racial disparities were not uniformly distributed but instead heavily concentrated in specific regions.

4.1.2 2022 Results

In 2022, disparities remained persistent but showed slightly compressed variability compared to 2021. The highest disparities were observed in:

- Virginia (0.9)
- Tennessee (0.9)
- Illinois (0.9)
- Michigan (0.9)
- North Carolina (0.9)

Although the maximum gap was lower than in 2021, the pattern of Southern and Midwestern concentration remained evident.

States such as Hawaii and Vermont again showed gaps near zero. Notably, the number of states with extreme disparities (>1.0) decreased relative to 2021.

Key observations for 2022

- Disparities persisted across many of the same states as 2021.
- Variability across states narrowed.
- Regional patterns remained consistent.

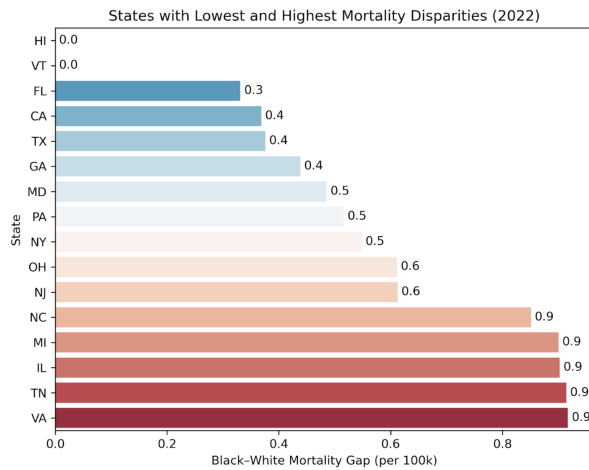


Figure 2: Ranked state-level maternal mortality gaps (2022), showing persistent disparities in multiple Southern and Mid-western states.

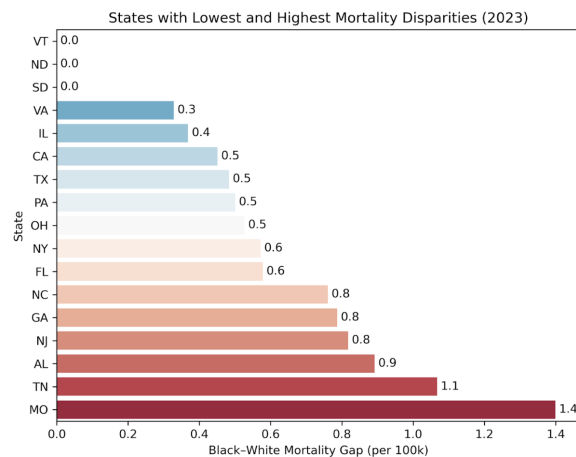


Figure 3: Ranked maternal mortality gaps (2023), highlighting states with largest racial disparities.

The 2022 results suggest that while extreme values may fluctuate annually, underlying structural inequities persist across years.

4.1.3 2023 Results

In 2023, the disparities again increased in select states. Missouri exhibited the largest gap (1.4), followed by Tennessee (1.1) and Alabama (0.9). North Carolina, Georgia and New Jersey also showed elevated disparities (0.8).

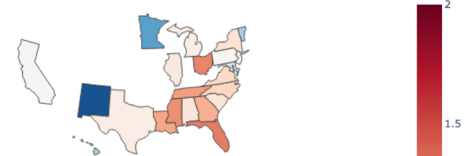
Several states reported zero disparities (Vermont, North Dakota, South Dakota), which may reflect small population counts, limited racial diversity, or statistical suppression in reporting. This is corroborated by looking at figure (Fig.3) where mortality ratios for these states are missing. This indicates a suppression at the CDC level of data that may have potentially misled gap statistics.

Key observations for 2023:

- Missouri emerged as the highest-disparity state.

Black-White Maternal Mortality Gap by Year

2021 - Black-White Maternal Mortality Gap (per 100k)



2022 - Black-White Maternal Mortality Gap (per 100k)



2023 - Black-White Maternal Mortality Gap (per 100k)

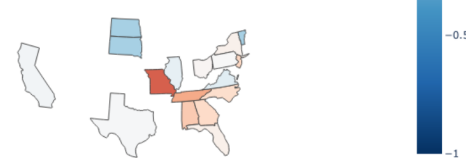


Figure 4: Maternal mortality gap by state (2021–2023). The ratio is calculated as previously reported. Florida reports the highest ratio in 2021, Illinois in 2022, and Missouri in 2023.

- Tennessee remained persistently high across all three years.
- Southern states continued to dominate the upper end of the ranking.
- Variability increased again relative to 2022.

The recurrence of certain states in the highest-disparity group across multiple years suggests that these disparities are not random fluctuations but may reflect enduring structural inequities.

4.2 Geographic Variation in Disparities

4.2.1 Southern Concentration of Disparities

Across all three years, the largest disparities are consistently concentrated in the Southeastern United States, including Tennessee, Mississippi, Alabama, Georgia, Louisiana, and Florida. These states frequently rank among the highest in both absolute gap and relative ratio measures. Southern states also tended to lack access to EHR interoperability (Fig. 7(b)), however, these same states more often have emergency services when compared with the same years' interoperability status (Fig. 7(a)).

This clustering suggests that disparities may be shaped by structural and policy-related factors, including healthcare access, Medicaid coverage differences, rural hospital availability, and broader social determinants of health.

Black / White Maternal Mortality Ratio by Year

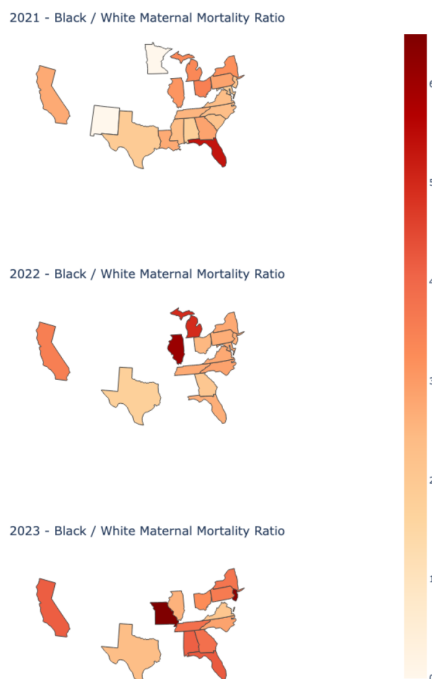


Figure 5: Maternal mortality ratio by state (2021–2023). The ratio is calculated as previously reported. Florida reports the highest ratio in 2021, Illinois in 2022, and Missouri in 2023.

Average Hospital Ratings by State (All Years)

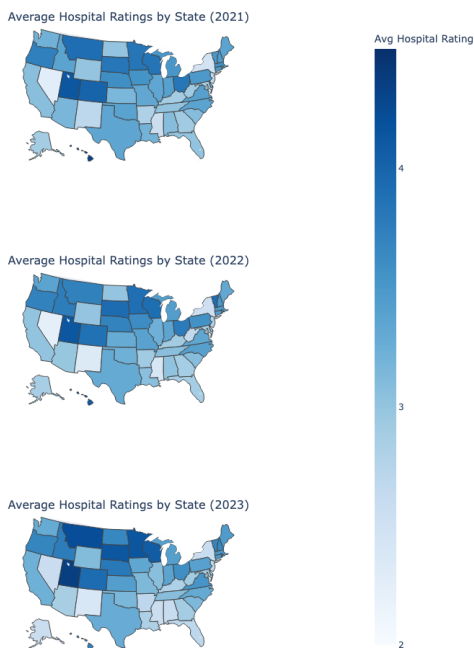
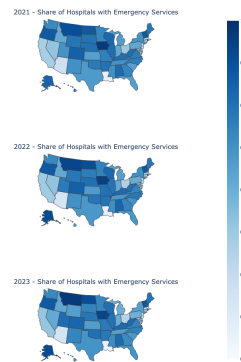


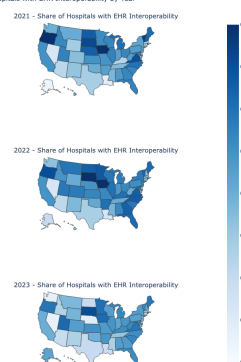
Figure 6: Average CMS hospital ratings by state, 2021–2023. Ratings show moderate geographic variation but do not clearly align with maternal mortality disparities.

Share of Hospitals with Emergency Services by Year



(a) Percentage of hospitals offering emergency services by state (October snapshots, 2021–2023).

Share of Hospitals with EHR Interoperability by Year



(b) Percentage of hospitals with EHR interoperability by state (October snapshots, 2021–2023).

Figure 7: Hospital infrastructure measures by state for October snapshots (2021–2023). Emergency service availability remains high and stable across years, while EHR interoperability shows variation across states.

4.2.2 Midwestern and Northeastern Variation

Several Midwestern states (e.g., Missouri, Illinois, Michigan, Ohio) also exhibit elevated disparities, particularly in specific years. Northeastern states generally show moderate but persistent inequities. The presence of disparities in states with relatively high hospital coverage indicates that access alone does not eliminate racial inequities in maternal outcomes.

4.2.3 States with Low or Zero Disparities

Some states (e.g., Vermont, Hawaii, North Dakota, South Dakota) display low or near-zero gaps. However, these findings should be interpreted cautiously, as small Black maternal populations, or even suppressed White maternal deaths in the data in the case of states with gaps, but no mortality ratios, may result in unstable or suppressed estimates.

4.2.4 Ratio vs. Gap Interpretation

Differences between the mortality gap (absolute difference) and ratio (relative risk) highlight the importance of using both

measures. A state may show a modest absolute gap but a high ratio if overall mortality rates differ substantially between racial groups.

In specific cases, while states like Florida – 2022 and 2023 – have a much lower mortality gap, the ratio is higher than when compared to most other states. Results like these can be indicative that rates for both Black and White populations may be lower, however, the rates for disparaged people still far outweigh White mothers.

4.2.5 Overall Pattern

The geographic distribution of disparities is not random. Disparities remain regionally clustered and relatively stable across years, suggesting that structural conditions—not short-term variation—drive inequities in maternal mortality outcomes.

4.3 Hospital Characteristics and Disparities

The map of the share of hospitals with emergency services shows that most states report high emergency service availability, often exceeding 75–80% percent of hospitals (Fig. 7(a)). Despite this relatively uniform access, the maternal mortality disparity maps indicate substantial variation in Black–White mortality gaps across states. High emergency coverage does not consistently correspond to smaller disparities, suggesting that basic emergency capacity alone is insufficient to address racial inequities in maternal outcomes.

On the other hand, the map indicates that hospitals with EHR interoperability varies drastically, with some states having as little as 50% of hospitals with access to electronic health records, while others have upwards of 90% (Fig. 7(b)). Despite varying accessibility, it appears that this functionality has little relation to the functionality of the hospitals as a whole, as hospital rating and mortality rates do not correlate to these levels.

Similarly, the average hospital rating by state map reveals moderate geographic variation in overall hospital quality, with many states clustered between 3 and 4 stars (Fig. 6). However, there is no clear relationship between higher average ratings and lower maternal mortality disparities. States with comparatively strong ratings still experience elevated racial gaps, indicating that general hospital performance metrics may not reflect maternal-specific care quality or equity practices.

Taken together, these visualizations suggest that while hospital infrastructure and quality are important components of healthcare systems, they do not fully account for observed racial disparities in maternal mortality. Structural, systemic, and social determinants of health likely play a substantial role beyond measurable hospital characteristics.

5 Discussion

The visual analyses highlight persistent and geographically patterned racial disparities in maternal mortality across the United States from 2021–2023. Both the mortality gap and mortality ratio maps demonstrate that disparities are not randomly distributed; instead, they cluster regionally, particularly in parts of the Southeastern and Midwestern United States.

Ranked bar charts further emphasize that certain states repeatedly appear among the highest-disparity states across years, suggesting structural persistence rather than isolated annual fluctuations.

When considered alongside the hospital access visualizations, a key finding emerges: high levels of hospital emergency service availability do not consistently correspond to smaller racial mortality gaps. Most states report relatively strong emergency service coverage, yet disparities remain substantial in many of them. Similarly, average hospital rating maps show moderate variation in overall hospital quality, but these patterns do not align clearly with the distribution of maternal mortality disparities. States with comparatively strong average ratings still exhibit elevated Black–White gaps, while some lower-rated states show smaller disparities.

Together, these findings suggest that hospital infrastructure and general quality indicators, while important, are insufficient to explain racial inequities in maternal mortality. The persistence and regional concentration of disparities point toward broader systemic influences, including structural racism, differences in quality of obstetric care, continuity of prenatal and postpartum services, insurance coverage, and underlying social determinants of health. The results reinforce that improving maternal health equity likely requires interventions that extend beyond hospital access alone and address deeper structural and policy-level inequities within healthcare systems and communities.

6 Limitations

This analysis is subject to several important limitations related to data availability, measurement, aggregation, and interpretation. These constraints shape both the scope of the findings and the conclusions that can be drawn.

6.1 Data Suppression and Missingness in Maternal Mortality Records

Maternal mortality data from CDC WONDER are subject to suppression rules designed to protect privacy, particularly for states or subgroups with small counts. As a result, some state–year–race combinations are either missing or partially observed, which limits geographic coverage and reduces the number of states included in the final analytic dataset. Although aggregation across years mitigates some suppression effects, it does not fully eliminate bias introduced by systematically missing low-population states. Consequently, the results are more representative of states with larger populations and more stable reporting.

Additionally, suppressed values were treated as missing rather than imputed. This conservative choice avoids introducing artificial precision but reduces sample size and may under-represent variability in states with smaller Black populations.

6.2 Aggregation Effects and Ecological Inference

All analyses were conducted at the state level. Aggregation masks substantial within-state heterogeneity in maternal outcomes, hospital distribution, and access to care. Urban–rural

divides, regional healthcare deserts, and local policy differences cannot be observed within this framework.

As a result, the findings are subject to ecological fallacy: observed state-level associations between hospital access and maternal mortality disparities cannot be assumed to hold at the individual, hospital, or community level. The analysis describes population-level patterns rather than individual risk.

6.3 Temporal Alignment and Snapshot Assumptions

Hospital system data were drawn from October snapshots for each year (2021–2023) to ensure consistency and comparability. While this approach avoids inconsistencies arising from mid-year reporting changes, it assumes that hospital characteristics are relatively stable within each year. Short-term changes, temporary closures, or rapid expansions in services may not be captured.

Similarly, maternal mortality data represent annual outcomes, and data is aggregated end of year, and thus cannot be temporally linked to specific hospital conditions at the time of death. This temporal mismatch limits causal interpretation.

6.4 Measurement Constraints in Hospital Access Indicators

Hospital access variables—such as emergency service availability, EHR interoperability, and overall hospital ratings—are coarse proxies for maternal healthcare quality. These measures do not capture obstetric-specific services, maternal care protocols, staffing adequacy, cultural competency, or continuity of prenatal and postpartum care.

Furthermore, maternal mortality data do not indicate where deaths occurred (e.g., hospital type or facility), preventing direct linkage between outcomes and specific hospital characteristics. As such, hospital access metrics should be interpreted as contextual indicators rather than direct explanatory factors.

6.5 Limited Demographic Scope

Due to data availability and suppression constraints, the analysis focuses exclusively on Black or African American and White racial categories. Other racial and ethnic groups—including Indigenous, Hispanic, Asian, and multiracial populations—are not represented. This limitation restricts the inclusivity of the findings and underscores the need for improved data collection practices that support equitable analysis across all populations.

6.6 Descriptive Rather Than Causal Analysis

This study is intentionally descriptive. While visualizations highlight co-variation between maternal mortality disparities and hospital access measures, no causal claims are made. Confounding variables—such as socioeconomic status, insurance coverage, maternal age, comorbidities, and structural racism—were not modeled and may drive observed patterns.

Statistical inference, predictive modeling, and hypothesis testing were deliberately excluded to avoid overstating conclusions given the data limitations.

6.7 Ethical Risks of Interpretation

There is an inherent ethical risk in visualizing disparities at the state level, as such presentations may inadvertently stigmatize

states or populations. To mitigate this risk, the analysis emphasizes structural patterns rather than attribution of blame, avoids ranking states as “good” or “bad,” and contextualizes disparities within broader systemic inequities.

6.8 Summary of Implications

Together, these limitations reinforce the importance of cautious interpretation. While the findings reveal persistent racial disparities in maternal mortality and contextualize them within healthcare infrastructure, they should be viewed as a starting point for deeper investigation rather than definitive explanations. The results highlight where disparities exist—not why they exist—and point toward the need for more granular, equity-focused data and analyses. Further analysis must also allow for a greater representation of data, both in the collection stage from the CDC, and after this, in the analysis stage by looking at expanded data to fully understand and highlight disparities in smaller and less populous states. It may also be important to further explore and investigate other equitable analyzes, such as law, and regulation focused issues especially regarding the health system to understand disparities.

7 Ethical Considerations

This study examines racial disparities in maternal mortality alongside state-level hospital access characteristics using publicly available, aggregated datasets. While the data do not contain personally identifiable information, the subject matter involves sensitive outcomes related to pregnancy, race, and healthcare access. As such, ethical considerations related to privacy, representation, interpretation, and potential misuse of findings are central to this analysis.

7.1 Data Privacy and Protection

All data used in this project are derived from publicly accessible sources and are aggregated at the state and year level. The maternal mortality data were obtained from CDC WONDER, which applies cell suppression and disclosure avoidance techniques to protect individual privacy when death counts are small. Hospital data were sourced from publicly available CMS hospital general information files and contain no patient-level records.

To further respect privacy constraints, analyses were restricted to state-level summaries and avoided any attempt to infer outcomes at the hospital or individual level. Suppressed or missing values were retained as missing rather than imputed to prevent the inadvertent reconstruction of protected information.

7.2 Representation and Bias

Racial categories in the maternal mortality dataset are limited to those defined and released by the CDC, with this analysis focusing specifically on Black or African American and White populations due to data completeness constraints. This limitation necessarily excludes other racial and ethnic groups and may obscure disparities experienced by Hispanic, Indigenous, Asian, or multiracial populations.

Additionally, states with suppressed or incomplete mortality data are underrepresented in visualizations and analyses, which may bias observed geographic patterns. These omissions are explicitly documented and visually encoded (e.g., missing states shown in neutral colors) to avoid misleading interpretations.

Hospital access metrics (such as emergency services availability or EHR interoperability) reflect structural features of healthcare systems rather than quality of obstetric or maternal care specifically. These measures may be influenced by reporting practices, hospital size, or administrative characteristics, which should not be conflated with direct patient outcomes.

It is important to note that the CMS dataset only reports on hospitals that are registered with Medicare/Medicaid. Hospitals that are not registered with Medi-Cal or Medicare are often private, concierge, or specialized, high-end or niche providers that opt-out of government programs. Therefore, our analysis is limited to publicly accessible data sources.

7.3 Interpretation and Causal Inference

This analysis is exploratory and descriptive in nature. While associations between maternal mortality disparities and hospital access characteristics are examined, no causal claims are made. The absence of patient-level linkage between maternal deaths and specific hospitals precludes attributing mortality outcomes to particular healthcare facilities or hospital types.

Visualizations and summary statistics are presented with careful framing to emphasize correlation rather than causation. The analysis avoids normative judgments about individual states or healthcare systems and instead focuses on identifying broad structural patterns that may warrant further investigation.

7.4 Risk of Stigmatization and Misuse

Maps and rankings of states by mortality disparity or hospital access carry the risk of stigmatizing regions or populations if interpreted without appropriate context. To mitigate this risk, figures are accompanied by explanatory captions that highlight data limitations, suppression effects, and the structural nature of the indicators used.

The results are not intended for performance evaluation, policy enforcement, or punitive comparisons between states. Instead, they are positioned as a tool for understanding systemic inequities and informing future research, policy discussions, and data collection improvements.

7.5 Transparency and Reproducibility

All data extraction, cleaning, and transformation steps are documented in accompanying data extraction and cleaning files. Methodological choices—including the use of October hospital data as an annual snapshot, handling of missing values, and normalization decisions—are explicitly justified to ensure transparency and reproducibility.

By maintaining clear documentation and acknowledging limitations, this project aims to support responsible use of the findings and encourage critical engagement rather than definitive conclusions.

8 Conclusion

This project examined state-level racial disparities in maternal mortality in the United States from 2021 to 2023 and explored how these disparities co-vary with measures of hospital access and infrastructure. By integrating maternal mortality data from CDC WONDER with hospital system characteristics from publicly available CMS hospital information files, the analysis provides a multi-year, structural view of inequities in maternal health outcomes and healthcare access.

Across all three years, maternal mortality rates for Black or African American women were consistently higher than those for White women in most states with sufficient data coverage. While the magnitude of the Black–White mortality gap varied by state and year, the persistence of these disparities highlights the entrenched nature of racial inequities in maternal health. States with the largest gaps often remained among those with elevated disparities across consecutive years, suggesting that these patterns are not transient or driven solely by short-term fluctuations.

The hospital access data revealed comparatively smaller variation across states than the mortality metrics. Most states reported a high proportion of hospitals offering emergency services, and adoption of EHR interoperability increased modestly over time. However, these broadly high access levels did not correspond to equally uniform maternal outcomes. States with similar levels of hospital access frequently exhibited markedly different mortality gaps, underscoring that access alone—at least as measured at the state level—does not fully explain racial disparities in maternal mortality.

Taken together, these findings suggest that maternal mortality disparities are shaped by factors that extend beyond the presence of hospitals or emergency services. Structural inequities, differences in quality of care, systemic racism within healthcare delivery, social determinants of health, and variation in prenatal and postpartum care continuity likely play significant roles. The inability to directly link maternal deaths to specific hospitals further reinforces that state-level infrastructure metrics, while informative, are insufficient for isolating causal mechanisms.

Importantly, this analysis was intentionally descriptive rather than causal. The visualizations and summary statistics are designed to highlight geographic patterns and disparities without attributing blame to individual states or healthcare providers. Data suppression, aggregation, and reporting limitations—particularly for smaller states or populations—constrain the granularity of conclusions and emphasize the need for cautious interpretation.

Despite these limitations, the project demonstrates the value of integrating heterogeneous public datasets to contextualize maternal mortality disparities within broader healthcare systems. The combined use of choropleth maps, ranked comparisons, and temporal views helps surface patterns that may not be apparent from a single dataset alone. These visual tools can support further research, guide hypothesis generation, and inform discussions around equity-focused healthcare policy.

Future work could extend this analysis by incorporating additional covariates such as state-level maternal health policies, Medicaid expansion status, rural–urban hospital distribution,

or socioeconomic indicators. More granular data—such as county-level mortality, hospital service lines, or patient-level outcomes—would enable deeper examination of mechanisms driving disparities. Additionally, expanding the racial and ethnic scope of the analysis as data availability improves is critical for a more inclusive understanding of maternal health inequities. Further important factors important to research are public policy, population trends, and if possible a way to account for suppressed data by determining counts, or more complete data from the CDC would improve equity and research.

Ultimately, this project reinforces a central public health insight: persistent racial disparities in maternal mortality cannot be addressed through healthcare access alone. Meaningful progress will require coordinated efforts that address structural inequities, improve quality and continuity of care, and center equity in maternal health policy and practice.

9 References

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